Name of Department:- Computer Science and Engineering

1.	Subject Code:	TDM 881		Course Title:	**Disaster Management**	
2.	Contact Hours:	L: 2	T: -	P: -		
3.	Semester: VIII					

4. Pre-requisite: None

5. Course Outcomes: After completion of the course students will be able to

1- Study and investigate the various types of Hazards and disasters and create awareness in the community to effectively prevent and react to such incidents.
2- Investigate and analyze hazards, disasters and measure their interrelationships with the developing humanitarian activities for solving future disaster problems.
3- To study, analyze and build skills to respond to disasters and hazards with community participation for controlling climate change.
4- To develop skills by training disaster forces and communities for successful Disaster Risk Reduction.
5- Understand the Disaster Management Laws and Policies and effectively apply for prevention and building the Disaster management system.
6- Building robust reaction and response systems by Technological innovations and skill building.

6. Detailed Syllabus

UNIT	CONTENTS	Contact Hrs
Unit – I	**Introduction, Definitions and Classification:** Concepts and definitions - Disaster, Hazard, Vulnerability, Resilience, Risks Natural disasters : Cloud bursts, earth quakes, Tsunami, snow, avalanches, landslides, forest fires,diversion of river routes (ex. Kosi river), Floods, Droughts Cyclones, volcanic hazards/ disasters(Mud volcanoes): causes and distribution, hazardous effects and environmental impacts of naturaldisasters, mitigation measures, natural disaster prune areas in India, major natural disasters in India with special reference to Uttarakhand.Man-induced disasters: water logging, subsidence, ground water depletion, soil erosion,, release of toxic gases and hazardous chemicals into environment , nuclear explosions	**9**
Unit – II	**Inter-relationship between Disasters and Development** Factors affecting vulnerabilities, differential impacts, impacts of development projects such asdams, embankments, changes in land use etc. climate change adaption, relevance of indigenous knowledge, appropriate technology and local resources, sustainable development and its role in disaster mitigation, roles and responsibilities of community, panchayat raj institutions/urban local bodies,	**8**

	state, centre and other stake holders in disaster mitigation.	
Unit – III	**Disaster Management (Pre-disasterstage, Emergency stage and Post Disaster Stage)** 1. Pre-disaster stage (preparedness): Preparing hazard zonation maps, predictably/forecastingand warning, preparing disaster preparedness plans, land use zoning, preparedness through information, education and communication (IEC), disaster resistant house construction, population reduction in vulnerable areas, awareness 2. Emergency Stage: Rescue training for search & operation at national & regional level,immediate relief, assessment surveys 3. Post Disaster stage: Rehabilitation and reconstruction of disaster affected areas; urban disaster mitigation: Political and administrative aspects, social aspects, economic aspects, environmental aspects.	**9**
Unit – IV	**Disaster Management Laws and Policies in India** Environmental legislations related to disaster management in India: Disaster Management Act,2005; Environmental policies & programs in India- Institutions &nationalcenters for natural disaster mitigation: National Disaster Management Authority (NDMA):structure and functional responsibilities, National Disaster Response Force (NDRF): Rule andresponsibilities, National Institute Of Disaster Management (NIDM): Rule and responsibilities.	**8**
	Total	**34**

.

Text Books:

- M MSulphey,” Disaster Management”, PHI, 2016

Name of Department:- Computer Science and Engineering

1.	Subject Code:	TCS 822				Course Title:	**Mobile Applications Development**
2.	Contact Hours:	L:	3	T:	-	P:	-

3. Semester: VIII

4. Pre-requisite: Excellent Knowledge of JAVA programming and Database Management System

5. Course Outcomes: After completion of the course students will be able to

1. Identify and apply the key technological principles and methods for delivering and maintaining mobile applications,
2. Evaluate and contrast requirements for mobile platforms to establish appropriate strategies for development and deployment,
3. Develop and apply current standard-compliant scripting/programming techniques for the successful deployment of mobile applications targeting a variety of platforms,
4. Carry out appropriate formative and summative evaluation and testingutilising a range of mobile platforms,
5. Interpret a scenario, plan, design and develop a prototype hybrid and native mobile application,
6. Investigate the leading edge developments in mobile application development and use these to inform the design process.

6. Detailed Syllabus

UNIT	**CONTENTS**	**Contact Hrs**
Unit - I	**Getting started with Mobility** Mobility landscape, Mobile platforms, Mobile apps development, Overview of Android platform, setting up the mobile app development environment along with an emulator, a case study on Mobile app development	**9**
Unit - II	**Building blocks of mobile apps** App user interface designing – mobile UI resources (Layout, UI elements, Draw-able, Menu), Activity- states and life cycle, interaction amongst activities. App functionality beyond user interface - Threads, Async task, Services – states and life cycle, Notifications, Broadcast receivers, Telephony and SMS APIs Native data handling – on-device file I/O, shared preferences, mobile databases such as SQLite, and enterprise data access (via Internet/Intranet)	**8**
Unit – III	**Sprucing up mobile apps** Graphics and animation – custom views, canvas, animation APIs, multimedia – audio/video playback and record, location awareness, and native hardware access (sensors such as accelerometer and gyroscope)	**9**

Unit – IV	**Testing mobile apps** Debugging mobile apps, White box testing, Black box testing, and test automation of mobile apps, JUnit for Android, Robotium, MonkeyTalk	**9**
Unit – V	**Taking apps to Market** Versioning, signing and packaging mobile apps, distributing apps on mobile market place	**8**
	Total	**43**

Text/ Reference Books:

1. Jeff McWherter, Scott Gowell, "Professional Mobile Application Development",Wrox Publication.
2. "Mobile Application Development "Black Book, Dreamtech Press

Name of Department:- Computer Science and Engineering

1. Subject Code: TCS 851 Course Title: **Storage Networks**
2. Contact Hours: L: 3 T: - P: -
3. Semester: VIII

4. Pre-requisite: Knowledge of Database and Networking is required

5. Course Outcomes: After completion of the course students will be able to

1. Understand the different aspects of storage management
2. Describe the various applications of RAID
3. Compare and contrast the I/O Techniques
4. Categorize virtualization on various levels of storage network
5. Estimate the various requirements of storage management systems
6. Design a complete data center and enhance employability in this field

6. Detailed Syllabus

UNIT	CONTENTS	Contact Hrs
Unit - I	Introduction to Storage Technology Introduction to storage network, Five pillars of IT, parameters related with storage, data proliferation, problem caused by data proliferation, Hierarchical storage management, Information life cycle management (ILM), Role of ILM, Information value vs. time mapping, Evolution of storage, Storage infrastructure component, basic storage management skills and activities, Introduction to Datacenters, Technical & Physical components for building datacenters	**10**
Unit - II	Technologies for Storage network Server centric IT architecture & its limitations, Storage centric IT architecture & advantages, replacing a server with storage networks, Disk subsystems, Architecture of disk subsystem, Hard disks and Internal I/O channel, JBOD, RAID& RAID levels, RAID parity, comparison of RAID levels, Hot sparing, Hot swapping, Caching : acceleration of hard disk access, Intelligent Disk subsystem architecture Tape drives: Introduction to tape drives, Tape media, caring for Tape& Tape heads, Tape drive performance, Linear tape technology, Helical scan tape technology	**9**
Unit – III	I/O techniques I/O path from CPU to storage systems, SCSI technology – basics & protocol, SCSI and storage networks, Limitations of SCSI	**10**

	Fibre channel: Fibre channel, characteristic of fibre channel, serial data transfer vs. parallel data transfer, Fibre channel protocol stack, Links, ports & topologies, Data transport in fibre channel, Addressing in fibre channel, Designing of FC-SAN, components, Interoperability of FCSAN, FC products IP Storage: IP storage standards (iSCSI, iFCP, FCIP, iSNS), IPSAN products, Security in IP SAN, introduction to InfiniBand, Architecture of InfiniBand NAS – Evolution, elements & connectivity, NAS architecture	
Unit – IV	Storage Virtualization Introduction to storage virtualization, products, definition, core concepts, virtualization on various levels of storage network, advantages and disadvantages, Symmetric and asymmetric virtualization, performance of San virtualization, Scaling storage with virtualization	**9**
Unit – V	Management of storage Networks Management of storage network, SNMP protocol, requirements of management systems, Management interfaces, Standardized and proprietary mechanism, In-band& Out-band management, Backup and Recovery	**8**
	Total	**46**

Text/ Reference Books:

1. "Storage Networks: The Complete Reference", R. Spalding, McGraw-Hill
2. "Storage Networking Fundamentals: An Introduction to Storage Devices, Subsystems, Applications, Management, and Filing Systems", Marc Farley, Cisco Press.
3. "Designing Storage Area Networks: A Practical Reference for Implementing Fibre Channel and IP SANs, Second Edition", Tom Clark Addison Wesley

Name of Department:- Computer Science and Engineering

1.	Subject Code:	TCS 855			Course Title:	**Agile Software Engineering**	
2.	Contact Hours:	L:	3	T:	-	P:	-
3.	Semester: VIII						

4. Pre-requisite: TCS 602

5. Course Outcomes: After completion of the course students will be able to

1. Describe two or more agile software development methodologies.
2. Identify the benefits and pitfalls of transitioning to agile.
3. Compare agile software development to traditional software development models.
4. Apply agile practices such as test-driven development, standup meetings, and pair programming to their software engineering practices.
5. Apply the agile testing
6. Describe the agile in current market scenario.

6. Detailed Syllabus

UNIT	**CONTENTS**	**Contact Hrs**
Unit - I	**Fundamentals of Agile:** The Genesis of Agile, Introduction and background, Agile Manifesto and Principles, Overview of Agile Methodologies – Scrum methodology, Extreme Programming, Feature Driven development, Design and development practices in an Agile projects, Test Driven Development, Continuous Integration, Refactoring, Pair Programming, Simple Design, User Stories, Agile Testing, Agile Tools	**10**
Unit - II	**Agile Project Management:** Agile Scrum Methodology, Project phases, Agile Estimation, Planning game, Product backlog, Sprint backlog, Iteration planning, User story definition, Characteristics and content of user stories, Acceptance tests and Verifying stories, Agile project velocity, Burn down chart, Sprint planning and retrospective, Daily scrum, Scrum roles – Product Owner, Scrum Master, Scrum Developer, Scrum case study, Tools for Agile project management	**10**
Unit – III	**Agile Software Design and Programming:**	**9**

	Agile Design Principles with UML examples, Single Responsibility Principle, Open Closed Principle, Liskov Substitution Principle, Interface Segregation Principles, Dependency Inversion Principle, Need and significance of Refactoring, Refactoring Techniques, Continuous Integration, Automated build tools, Version control, Test-Driven Development (TDD), xUnit framework and tools for TDD	
Unit – IV	**Agile Testing:** The Agile lifecycle and its impact on testing, Testing user stories - acceptance tests and scenarios, Planning and managing Agile testing, Exploratory testing, Risk based testing, Regression tests, Test Automation, Tools to support the Agile tester	**9**
Unit – V	**Agile in Market:** Market scenario and adoption of Agile, Roles in an Agile project, Agile applicability, Agile in Distributed teams, Business benefits, Challenges in Agile, Risks and Mitigation, Agile projects on Cloud, Balancing Agility with Discipline, Agile rapid development technologies	**8**
	Total	**46**

Text Book:

1. Ken Schawber, Mike Beedle, “Agile Software Development with Scrum”, Pearson, 2008

Name of Department:- Computer Science and Engineering

1. Subject Code: TOE 811 Course Title: **Mobile Computing**
2. Contact Hours: L: 3 T: - P: -
3. Semester: VIII

4. Pre-requisite: TIT 704, TCS 703

5. Course Outcomes: After completion of the course students will be able to

1. Explain the concepts, techniques, protocols and architecture employed in wireless local area networks, cellular networks, and Adhoc Networks
2. Describe and analyze the network infrastructure requirements to support mobile devices and users.
3. Interpret data management issues and distributed file system.
4. Asses the important issues and pertaining to clustering in wireless networks.
5. Value assessment of mobile agent in mobile computing environment.
6. Investigate Adhoc Routing Protocol.

6. Detailed Syllabus

UNIT	CONTENTS	Contact Hrs
Unit - I	Introduction, issues in mobile computing, overview of wireless telephony: cellular concept, GSM: air-interface, channel structure, location management: HLR-VLR, hierarchical, handoffs, channel allocation in cellular systems, CDMA, GPRS	**9**
Unit - II	Wireless Networking, Wireless LAN Overview: MAC issues, IEEE 802.11, Blue Tooth, Wireless multiple access protocols, TCP over wireless, Wireless applications, data broadcasting, Mobile IP, WAP: Architecture, protocol stack, application environment, applications	**8**
Unit – III	Data management issues, data replication for mobile computers, adaptive clustering for mobile wireless networks, CODA File system, Disconnected operations	**9**
Unit – IV	Mobile Agents computing, security and fault tolerance, transaction processing in mobile computing environment.	**8**
Unit – V	Ad Hoc networks, localization, MAC issues, Routing protocols, global state routing (GSR), Destination sequenced distance vector routing (DSDV), Dynamic source routing (DSR), Ad Hoc on demand distance vector routing (AODV), Optimized link state routing protocol (OLSR), QoS in Ad Hoc Networks, applications	**9**
	Total	**43**

.

Text/ Reference Books:

1. D.P. Agarwal, Qing Amazing"Introduction to wireless and Mobile systems" , Cengage learning India
2. J. Schiller," Mobile Communications", Addison Wesley.
3. Raj Pandya "Mobile and personal communication systems and services" IEEE press.
4. Kukumgarg , "Mobile computing – Theory and practice ", pearson.